

ENTANGLEMENT WITHOUT SPOOKINESS

Shared origin, product structure, and correlation without a travelling signal

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ONE COMPOSITE STATE

8/15 -> 1/15; shadows (2,3) -> (1,1)

Correlation is inherited from a shared whole; no third signal channel is generated.

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Abstract

Smithian Fold Theory derives entanglement as the evolution of one composite state rather than communication between two independent objects. From one machine-checked self-proven theorem, zero axioms and zero fitted parameters, the binary and colour periods 2 and 3 are coprime and generate one joint period 6, larger than their sum. A second route forms the shared dimension $3 \times 5 = 15$, places the composite at the single origin $1/15$, and verifies that one fold advances the same connected orbit to $2/15$.

The complete composite certificate makes the relation explicit. The state $8/15$ has local shadows (2,3) in the three- and five-worlds; one fold sends the whole to $1/15$, and both shadows reach (1,1) in that same operation. Each local law agrees with the global fold. No signal has to cross a separation because the correlation is inherited from the shared whole.

The no-extra-channel certificate closes the signalling question. The wave and structural channels are $2/3$ and $1/3$; they partition the One, fold into each other, and their difference remains inside the same two-member cycle. No third carrier is generated. The result supplies an exact, inspectable construction for nonlocal correlation and no signalling. It is not an opaque prediction presented as an explanation.

Central result

Layer	Exact result
Foundation	One self-proven theorem; zero axioms; zero fitted parameters
Primitive periods	2 and 3
Coprimality	$\gcd(2,3)=1$
Joint period	$\text{lcm}(2,3)=6$
Joint surplus	$6 > 2+3$
Shared dimension	$3 \times (3+2)=15$
Shared origin	$1/15$
Connected advance	$\text{fold}(1/15)=2/15$
Composite walk	$8/15 \rightarrow 1/15$
Local shadows	$(2,3) \rightarrow (1,1)$ together
Channel partition	$2/3 + 1/3 = 1$
Channel closure	fold and difference remain within the two-member cycle

1. One state, two readings

Entanglement is often narrated as one particle acting on another across a distance. SFT changes the object before explaining the correlation: the entangled pair is not constructed as two independent states plus a later link. It is one composite state with two local readings.

The distinction is physical. If the global object is one, a fold of the whole updates both readings without transmitting an instruction between them. Correlation is inherited from common structure, not added by a travelling mechanism.

2. Product structure from the generators

The smallest forced subsystem periods are the binary and colour counts:

$a=2, b=3$.

They are coprime, so their least common multiple is their product:

$\text{lcm}(2,3)=2 \times 3=6$.

The joint period exceeds the sum of the separate periods:

$$6 > 2+3=5.$$

The composite therefore contains a relation that cannot be represented by simply listing the two separate cycles. This exact product surplus is the engine's structural form of entanglement.

3. The shared origin

The nonlocal-correlation engine takes the colour count three and the next odd count five. Their coprime product gives the joint dimension

$$D=3*5=15.$$

The composite begins at one shared state

$$x=1/15.$$

Applying the fold advances it on the same orbit:

$$\text{fold}(1/15)=2/15.$$

There is one numerator evolving in one joint denominator. The two subsystem descriptions are projections of that connected orbit, not independently evolving local stores.

4. The complete composite walk

The entangled-composites engine starts at 8/15. Its shadows are

$$8 \bmod 3 = 2, \text{ and } 8 \bmod 5 = 3.$$

One fold doubles the global numerator:

$$2*8=16=1 \bmod 15,$$

so the whole lands at 1/15. The new shadows are

$$1 \bmod 3 = 1, \text{ and } 1 \bmod 5 = 1.$$

Both local readings therefore reach their origin together. The local dynamics agree:

$$2*2=1 \bmod 3, \text{ and } 2*3=1 \bmod 5.$$

The global and local accounts are identical descriptions of the same operation.

5. Why no signal is available

The quantum-communication engine derives two channel shares:

- wave channel 2/3;
- structural channel 1/3.

They partition the One. Under the fold they exchange:

$$\text{fold}(1/3)=2/3, \text{ and } \text{fold}(2/3)=1/3.$$

Their difference is 1/3, which is already the structural channel, and folding it returns the wave channel. The complete construction remains inside one closed two-member cycle. There is no generated third channel on which an additional controllable message can ride.

Nonlocal correlation and no signalling are therefore simultaneous consequences of one object: the whole resolves globally, while its channel census supplies no independent signal carrier.

6. Bell evidence and the fold

Bell experiments exclude local factorizations that assign independent hidden answers to separated parts. SFT does not construct such a factorization. Its starting object is the shared composite and its single fold orbit. The observed nonlocal correlations are therefore evidence for the kind of global structure the engine represents, not evidence that mathematical determination must be abandoned.

The claim is precise: correlation is nonlocal, the forward law is deterministic, and the closed channel census prevents controllable faster-than-light signalling.

7. What makes this empirical proof rather than opaque reliability

The paper does not ask a black box to guess correlations and then call accuracy an explanation. It exposes the complete chain:

1. the theorem forces the generating counts;
2. coprimality forces the product period;
3. exact fractions construct one shared origin;
4. the fold advances the composite and its local shadows together;
5. the channel cycle is exhaustively closed;
6. machine certificates reproduce every equality;
7. experiment compares the declared global correlations and absence of a signal channel with nature.

Reliable output is evidence of performance. Transparent derivation plus execution plus measurement is evidence of structure. The latter is the empirical standard used here.

8. Reproduction

```
./verify/test_entanglement
./verify/test_nonlocal_correlation
./verify/test_quantum_communication
./verify/test_entangled_universes
```

The current focused run returns every registered relation as ok. Source modules, test expressions and generated C certificates are all public in the synchronized corpus.

9. Empirical frontier

The current derivation opens direct work: scale coprime composites and their projected shadows; derive measurement settings from fold-orbit positions and compare Bell tables with exact shared-origin counts; and test two-channel closure in communication protocols. The theorem contains no wall against those extensions. The current state proves the finite product, shared-origin, synchronized-shadow and channel-closure relations; the next state scales the same transparent construction into fuller experimental tables.

Conclusion

Entanglement is not spooky action in SFT. It is one product state with a shared origin, one connected orbit and two local shadows. A single fold updates the whole and both shadows together, while the two generated channels close to the One and admit no extra signalling carrier. The construction is deterministic, nonlocal, no-signalling, exact and machine-checkable.

References

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